

Analysis General Exam, January 2011

January, 2011

Closed book, closed notes. Please pledge. In each problem, justify all assertions, show calculations, and identify those theorems which you invoke in your arguments.

Problem 1

Let $\{f_n\}$ be a sequence of real-valued continuous functions on $[0, 1]$ which is monotone non-increasing $f_{n+1}(x) \leq f_n(x)$ for all $x \in [0, 1]$ and such that

$$\lim_{n \rightarrow \infty} f_n(x) = 0$$

- a) Prove that the convergence is uniform.
- b) Show that, if instead $\{f_n\}$ is again a monotone sequence of continuous converging pointwise to a function f which is however not continuous, then the convergence is not uniform.

Problem 2

Let $\{f_n\}$ be a sequence of real-valued Borel measurable functions on $\mathbb{R}$.

- a) Show that

$$f(x) \equiv \sup_n f_n(x)$$

and

$$g(x) \equiv \limsup f_n(x)$$

are measurable.

- b) Define the set K ,

$$K = \{x : f_n(x) \in (0, 1) \text{ for infinitely many } n\text{'s}\}.$$

Show that K is a Borel measurable set.

Problem 3

Let m be Lebesgue measure, and suppose that $f(x, y)$ is a Lebesgue measurable non-negative function on the plane $\mathbb{R}^2$ such that

$$F(\lambda, y) = m\{x : f(x, y) \geq \lambda\}$$

satisfies

$$\int_0^\infty \int_{\mathbb{R}} \lambda^r F(\lambda, y) dy d\lambda < \infty$$

for some $r \geq 0$.

Let

$$G(\lambda, x) = m\{y : f(x, y) \geq \lambda\}$$

a) Show that

$$\int_0^\infty \int_{\mathbb{R}} \lambda^r G(\lambda, x) dx d\lambda < \infty$$

b) Show that $f \in L^{r+1}(\mathbb{R}^2, dxdy)$, i.e.,

$$\int_{\mathbb{R}^2} f^{r+1}(x, y) dxdy < \infty$$

Show also that

$$m \times m \{(x, y) \in \mathbb{R}^2 : f(x, y) \geq \lambda\} \leq \frac{c}{\lambda^{r+1}}$$

with $m \times m$ Lebesgue measure on the plane and with c a finite constant.

Problem 4

Let $\{f_n\}$ be the sequence of functions defined on $[0, 2\pi]$ with

$$f_n(x) = \sum_{k=1}^n \frac{e^{ikx}}{k^{3/4}}$$

a) Show that $\{f_n\}$ converges in an $L^2([0, 2\pi], dx)$ -sense, $n \rightarrow \infty$.

b) Show that $\{f_n\}$ converges in an $L^1([0, 2\pi], dx)$ -sense, $n \rightarrow \infty$.

Problem 5

Using residue methods, find

$$\int_{-\infty}^{\infty} \frac{\cos x}{e^x + e^{-x}} dx$$

by considering

$$\int_{\Gamma} \frac{e^{iz}}{e^z + e^{-z}} dz$$

where Γ is the rectangle as shown with a suitably chosen value for the height.

Problem 6

Suppose that f is analytic in an open connected set Ω , and that all values of f on Ω lie in the disk of radius $M > 0$ centered at 0. Prove that

$$(*) |f'(z)| \leq \frac{M}{d(z)}$$

for all $z \in \Omega$, where $d(z)$ is the distance from z to the boundary of Ω . Then show that $(*)$ can be used to prove Liouville's theorem.

Problem 7

Suppose f is analytic in a set containing the closed unit disk $\overline{\mathbb{D}} = \{z : |z| \leq 1\}$ with $f(-\log 2) = 0$ and $|f(z)| \leq |e^z|$ for all z with $|z| = 1$. How large can $|f(\log 2)|$ be? (Here, $\log z$ denotes the principal branch of the logarithm.)

Problem 8

a) Find the image of the unit disk $\mathbb{D} = \{z : |z| < 1\}$ under the mapping

$$g(z) = \frac{z+1}{1-z}$$

b) Find the image of all straight lines through the point $z = 1$ under this mapping.

c) Show that the function

$$f(z) = e^{-g(z)}$$

is bounded on the unit disk. Determine the limit of $f(z)$ as $z \rightarrow 1$ along any line segment lying within the unit disk. What is the limit as $z \rightarrow 1$ along the unit circle $\{z : |z| = 1\}$?